WI-FI NETWORK SECURITY ANALYSIS

Test 2. Secure Infrastructure Configurations – Exercise 6

Code: Test 2_Exercise 6
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Introduction

The goal of this exercise is to check the security level of the Wi-Fi network my computer is connected to, before and after changing the router's Wi-Fi password.

The exercise required three steps:

1. Scan the Wi-Fi network with a network scanning tool and record its security level.
2. Change the Wi-Fi password of the router to a strong password.
3. Scan the network again with the same tool and compare the results.

Tool Used

Acrylic Analyzer, a Wi-Fi scanning tool that shows the networks in range and rates each network with a Network Quality score from 0 to 10.

Step 1: Initial Scan

- I scanned the Wi-Fi network my computer is connected to.
- The network name (SSID) is: DIGIFIBRA-PLUS-XXXX
- Network Quality: **6/10**
- The current password was weak: 12 characters, no symbols.

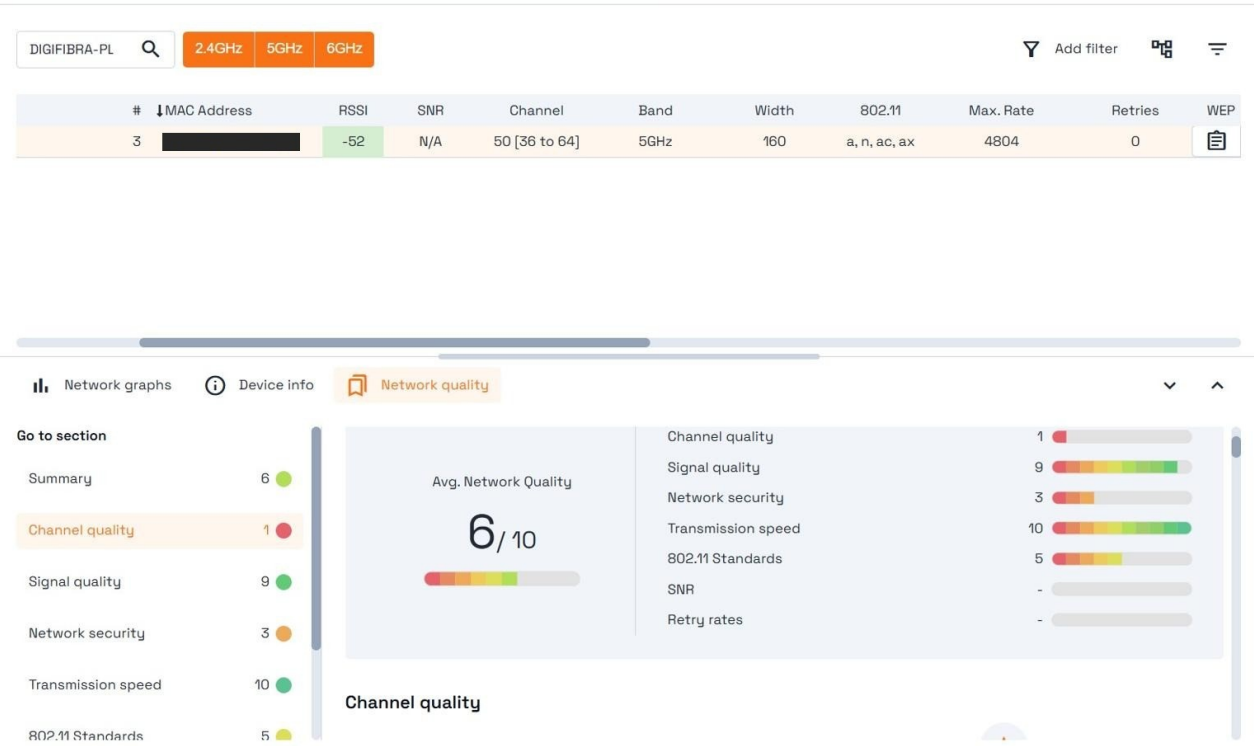


Figure 1: Network_Quality_Before

Step 2: Password Change

- I accessed the router settings from the browser, using the IP address 192.168.1.1.
- I changed the password to a stronger one: 20 characters, including uppercase letters, lowercase letters, numbers and symbols.

Step 3: Second Scan

- I scanned the network again with the same tool.
- Network Quality: **7/10**

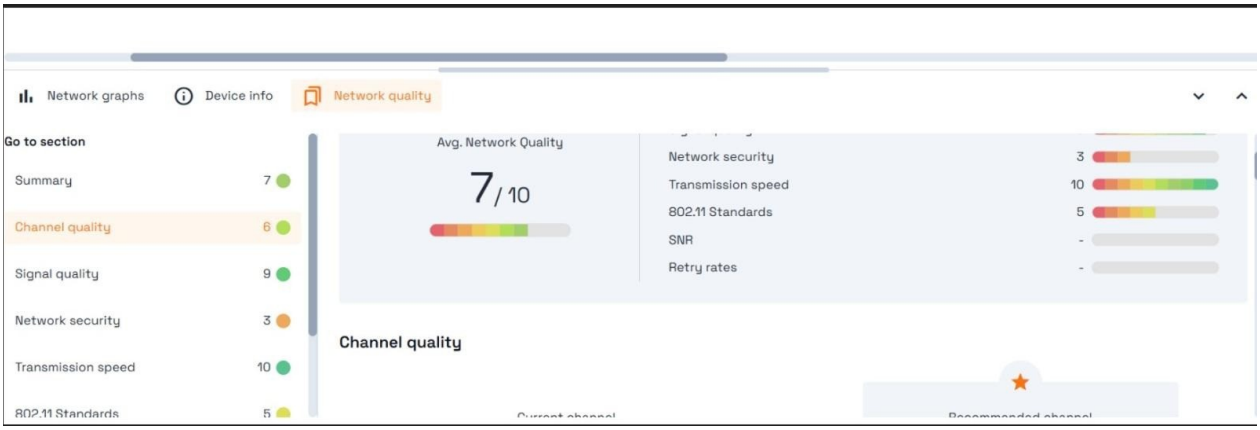


Figure 2: Network_Quality_After

Results

The overall score increased from 6/10 to 7/10, due to the improvement in channel quality. The Network security score remained 3/10, because the tool evaluates the encryption protocol announced by the router, not the password.

Conclusion

The strong password (20 characters, with uppercase letters, lowercase letters, numbers and symbols) improves the protection against brute-force attacks and unauthorised access.